

Representational Measurement of Similarity: The Additive Difference Model, Revisited

Arthur Liske, Mathematik

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Abstract text.....

Contents

1	Introduction	1
2	Proximity Structures and Metrics	1
3	Segmental Additivity	3
3.1	Segmentally Additive Metrics	3
3.2	Segmentally Additive Proximity Structures	9
	seg add representability	10
4	Additive Difference Models and Structures	11
4.1	Additive Difference Model	11
4.2	Additive Difference Structure	11
5	Combining Segmental Additivity with the AD-Model	11
5.1	Preliminary lemma	11
	Theorem name	20
6	Discussion	23

1 Introduction

2 Proximity Structures and Metrics

In diesem Kapitel geht es erstmal darum die Mathematische Basis Aufzubauen. Wie aus der Einleitung hervorgeht, wollen wir jeweils zwei Paare von Stimuli auf ihre Ähnlichkeit vergleichen. Seien a, b, c und d solche Stimuli. Dann bedeutet die Notation $(a, b) \succsim (c, d)$, dass Stimuli a und b sich unähnlicher sind, als c und d . Analog bedeutet $\sim$, dass die Ähnlichkeit zweier Paare gleich ist.

Die Idee von einer proximity structure, ist die Mathematische Definition dieser Idee: eine Relation $\succsim$ auf einer Menge A die die Grundeigenschaften einer Metrik erfüllt.

Definition 2.1 (factorial proximity structure).

Suppose A is a set and $\succsim$ a quaternary relation on A , i.e., binary relation on $A \times A$. $\langle A, \succsim \rangle$ is a proximity structure iff the following axioms hold for all $a, b \in A$:

1. $\succsim$ is a weak ordering on $A \times A$, i.e., it is connected and transitive.
2. $(a, b) \succsim (a, a)$ whenever $a \neq b$.
3. $(a, a) \sim (b, b)$ (minimality).
4. $(a, b) \sim (b, a)$ (symmetry).

The structure is factorial iff

$$A = \prod_{i=1}^n A_i.$$

Remark 2.2. Zusammenhängend heißt, man kann alle distinkten Elemente vergleichen, also für eine Relation R auf einer Menge X gilt für $x, y \in X, x \neq y$ dann xRy oder yRx .

Da wir die Relation $\succsim$ durch eine Metrik repräsentieren wollen, lohnt es sich einen Begriff dafür einzuführen. Die Idee ist das, was man sich intuitiv vorstellt: man will dass die Metrik die Struktur von $\langle A, \succsim \rangle$ erhält, und die einzige Struktur die wir gegeben haben, ist die Ordnung jeweils zweier Paare.

Definition 2.3 (Representability of $\succsim$ by a metric).

A proximity structure $\langle A, \succsim \rangle$ is representable by a metric iff there exists a real-valued function δ on A such that:

1. $\langle A, \delta \rangle$ is a metric space, i.e., δ is a metric on A .
2. δ preserves the relation $\succsim$, i.e., for all $a, b, c, d \in A$

$$\delta(a, b) \geq \delta(c, d) \iff (a, b) \succsim (c, d).$$

3 Segmental Additivity

3.1 Segmentally Additive Metrics

Da Metriken an sich wenig Struktur bieten, lohnt es sich noch eine Eigenschaft mehr zu fordern.

Diese Eigenschaft ist segmental additivity und beschreibt, dass es zu zwei Punkten immer eine Kurve gibt, die diese verbindet und die Abstände additiv sind, das heißt auf dem Segment die Dreiecksungleichung mit Gleichheit gilt. Für uns reicht es zu wissen, dass wenn b auf dem Segment zwischen a und c liegt, dass dann $\delta(a, b) + \delta(b, c) = \delta(a, c)$ gilt. Für die genaue Definition siehe Anhang Am Ende dieses Abschnittes sind Beispiele für Metriken mit additiven Segmenten angegeben.

Der Folgende Satz liefert eine (bis auf positive Skalierung) eindeutige Metrik die die Relation repräsentiert und zusätzlich zur Ordnung, die Eigenschaft segmental solvability erhält.

Theorem 3.1 (Representation theorem for segmentally additive proximity structures). *Suppose $\langle A, \succsim \rangle$ is a segmentally additive proximity structure. Then $\langle A, \succsim \rangle$ is representable by a metric, and additionally, for all $a, b, c \in A$:*

1. $\langle abc \rangle \iff \delta(a, b) + \delta(b, c) = \delta(a, c)$.
2. *If δ' is another metric on A that satisfies the above conditions, then there exists $\alpha > 0$ such that $\delta' = \alpha \delta$ (δ is a ratio scale).*

Definition 3.2.

Todo: Metric, open, convex, stetig ... definieren?

The definitions and statements from this section are taken out from (Busemann, 1955).

Definition 3.3 (curves and segments).

1. Let δ be a metric on X . A parametric curve in $\langle X, \delta \rangle$ is a function γ that maps a closed interval $[a, b] \subset \mathbb{R}$ into X and is continuous. The parametric curve γ is said to join $\gamma(a)$ and $\gamma(b)$.
2. Let γ be a parametric curve in $\langle X, \delta \rangle$ with domain $[a, b]$. Its length $L(\gamma)$ is the supremum of the sums

$$\sum_{i=1}^n \delta[\gamma(a_{i-1}), \gamma(a_i)]$$

over all finite sequences $a_0, a_1, \dots, a_n$ such that

$$a = a_0 \leq a_1 \leq \dots \leq a_n = b.$$

If $L(\gamma) < \infty$, then γ is rectifiable.

3. A rectifiable curve is a segment iff its length is equal to the distance between its endpoints, i.e. $L(\gamma) = \delta(a, b)$.
4. For $a \leq x \leq b$ we denote the length of the sub-curve $\gamma(t)$, $a \leq t \leq x$ from a to x by $L_a^x(\gamma)$. This is also called the *arc length* of the curve.

Remark 3.4 (continuity & monotonicity of sub length). For a rectifiable curve $\gamma(t)$, $a \leq t \leq b$, the arc length $L_a^x(\gamma)$ is a continuous, non-decreasing function of x .

Definition 3.5 (Metric space with additive segments).

Let δ be a metric on X . $\langle X, \delta \rangle$ is a metric space with additive segments iff for any $x, y \in X$ there is a segment joining them.

Example 3.6 (Minkowski bzw. l^p bzw. power metric).

Die l^p Metrik, die durch $\delta(a, b) = [\sum_{i=1}^n |\varphi_i(a_i) - \varphi_i(b_i)|^p]^{1/p}$ definiert ist, ist für $p \geq 1$ eine Metrik mit additiven Segmenten. Die Segmente sind genau die Verbindungslinien zwischen zwei Punkten, also alle Geraden.

Proof. Seien $x, z \in \mathbb{R}^n$ und $0 \leq t \leq 1$. Dann gilt für $y = (1-t)x + tz$ (y ist genau dann auf dem Geradensegment von z zu x , wenn man y so darstellen kann):

$$\begin{aligned} d(x, y) + d(y, z) &= \left[\sum_{i=1}^n |x_i - (1-t)x_i - tz_i|^p \right]^{1/p} + \left[\sum_{i=1}^n |(1-t)x_i + tz_i - z_i|^p \right]^{1/p} \\ &= t \left[\sum_{i=1}^n |x_i - z_i|^p \right]^{1/p} + (1-t) \left[\sum_{i=1}^n |x_i - z_i|^p \right]^{1/p} \\ &= d(x, z). \end{aligned}$$

□

Remark 3.7 (additive difference model). Es fällt auf, dass die l^p Metrik eine besondere Form hat, die durch $\delta(x, y) = F^{-1} [\sum_{i=1}^n F(|x_i - y_i|)]$ gegeben ist. Man kann diese Form auf 4 Stufen abstrahieren:

1. Decomposability: Die Distanz ist eine Funktion von Beiträgen die Komponentenweise sind.

$$\delta(a, b) = F[\psi(a_1, b_1), \dots, \psi(a_n, b_n)] \quad (1)$$

2. Intradimensional subtractivity: Die Komponentenweise Beiträge sind Beträge der Differenzen in den Dimensionen.

$$\delta(a, b) = F[|\varphi_1(a_1) - \varphi_1(b_1)|, \dots, |\varphi_n(a_n) - \varphi_n(b_n)|] \quad (2)$$

3. Interdimensionale additivity: Der Abstand ist eine Summe der Komponentenweise Beiträge.

$$\delta(a, b) = G \left[\sum_{i=1}^n \psi_i(a_i, b_i) \right] \quad (3)$$

4. additive difference model: Wenn alle 3 Eigenschaften gelten.

$$\delta(a, b) = G \left\{ \sum_{i=1}^n F_i[|\varphi_i(a_i) - \varphi_i(b_i)|] \right\} \quad (4)$$

Falls die Elemente a, b aus der proximity structure sind, sind noch Folgende Eigenschaften von den Funktionen gefordert:

1. F ist steigend in allen Argumenten
2. ψ_i ist für alle $i = 1, \dots, n$ reellwertig, symmetrischen und erfüllt $\psi_i(a_i, b_i) > \psi_i(a_i, a_i)$ wenn $a_i \neq b_i$
3. ϕ_i ist reellwertig für alle $i = 1, \dots, n$
4. G und F_i sind steigend ($i = 1, \dots, n$)

Man sieht, dass die l^p Metrik ein Spezialfall des additive difference models ist. Von besonderem Interesse sind Modelle wo die F_i alle über eine gemeinsame Funktion F (bis auf Streckung/Stauchung des Definitionsbereiches, $F_i(x) = F(t_i x)$, $t_i > 0$) dargestellt werden können und $G = F^{-1}$ gilt. Diese Bedingungen erfüllt die l^p Metrik auch. Hier kommt ein weiteres Beispiel für eine solche Metrik.

Example 3.8 (p-exp Metrik Familie).

Sei das F im additive difference model (9) gegeben durch

$$F(t) = q \cdot (e^{t \cdot r} - 1)^p \quad q, r > 0, p \geq 1 \quad (5)$$

Dann ist das Modell eine Metrik.

Proof. 1. Positivität: $\delta(a, b) = 0$ also $F(|a_i - b_i|) = 0$ für alle i , das ist nur der Fall wenn der Exponent Null ist, also $a_i = b_i$ gilt für alle i .

Anderes herum gilt, wenn $a \neq b$ ist, dass dann $a_i \neq b_i$ für mindestens ein i , woraus folgt dass $F_i(|a_i - b_i|) > 0$, also auch $\delta(a, b) > 0$.

2. Symmetrie: Symmetrie kommt daher, dass $|a_i - b_i| = |b_i - a_i|$ gilt.

3. Dreiecksungleichung: Die Dreiecksungleichung kommt aus dem Satz von Muhlolland **mulholland**. Das Paper von Mulholland ist auf Moodle zu finden.

□

Theorem 3.9 (Mulholland).

Die Ungleichung

$$f^{-1} \left[\sum_i f(|x_i + y_i|) \right] \leq f^{-1} \left[\sum_i f(|x_i|) \right] + f^{-1} \left[\sum_i f(|y_i|) \right].$$

gilt für alle $x, y \in \mathbb{R}^n$, $n = 1, 2, \dots$ wann immer die folgenden drei Bedingungen erfüllt sind:

1. f ist stetig und streng monoton steigend, und $f(0) = 0$.
2. f ist konvex auf $[0, \infty)$.
3. $\log f(e^x)$ ist konvex für $x \in (-\infty, \infty)$.

Beweis, dass 3.8 die Bedingungen von Mulholland erfüllt.

Zu Punkt 1: Stetigkeit und $f(0) = 0$ ist klar. Für Monotonie gilt, $e^{rt} - 1$ ist steigend, Potenzieren auch und multiplizieren mit $q > 0$ auch, damit ist die Komposition davon auch steigend. Zum Punkt 2: Selbes Argument wie bei Monotonie, nur mit Konvexität und Komposition von konvexen Funktionen ist wieder konvex.

Zum dritten Punkt, dass $\log F(e^t)$ konvex ist. Es gilt

$$\log q(e^{re^x} - 1)^p = \log q + p \cdot \log(e^{re^x} - 1)$$

Addition und Multiplikation mit $p \geq 1$ erhält Konvexität, also reicht es Konvexität von $\log(e^{re^x} - 1)$ zu zeigen.

Das zeigen wir über die zweite Ableitung, denn wenn die zweite Ableitung strikt positiv ist, ist die Funktion konvex.

Zweifaches Ableiten liefert

$$\frac{re^{re^x+x} (r(-e^x) + e^{re^x} - 1)}{(e^{re^x} - 1)^2} \stackrel{!}{>} 0$$

Wir sehen der Nenner ist immer positiv und ungleich 0, das heißt wir können den Nenner weg lassen. Genauso ist der Faktor re^{re^x+x} immer positiv und kann somit auch weggelassen werden. Es bleibt also zu zeigen, dass

$$\begin{aligned} -re^x + e^{re^x} - 1 &> 0 \\ \iff e^{re^x} - re^x &> 1 \end{aligned}$$

gilt.

Dies kann man in 2 Schritten zeigen: 1. die Linke Seite ist streng monoton steigend und 2. der Grenzwert für $x \rightarrow -\infty$ ist 1. Somit gilt die Ungleichung für alle $x \in \mathbb{R}$.

1. Für die Ableitung gilt:

$$re^x \cdot e^{re^x} - re^x = re^x \cdot (e^{re^x} - 1) > 0$$

da $e^x > 0$ gilt.

- 2.

$$\lim_{x \rightarrow -\infty} e^{re^x} - re^x = 1 - 0 = 1$$

Damit ist die Konvexität gezeigt, und die Behauptung folgt aus dem Satz von Mulholland.

□

Remark 3.10 (l^p Metrik $\in$ p-exp Metrik Familie).

Wenn wir in (5) für $q = r^{-p}$ einsetzen, erhalten wir im Grenzwert für $r \rightarrow 0$ die l^p Metrik. Das ist von Bedeutung, da dies mithilfe des Hauptresultates 5.13 zeigt, dass die Metriken der p-exp Familie keine Metriken mit additiven Segmenten sein können, obwohl man die l^p Metrik durch Grenzwertbildung aus dieser Familie darstellen kann. Das bedeutet, dass im Grenzwertprozess aus der Eigenschaft "additiv auf den Koordinatenachsen", die Eigenschaft "additiv auf beliebigen Liniensegmenten" wird.

Proof. Setzt man $q = r^{-p}$ erhält man:

$$r^{-p}(e^{rt} - 1)^p = \left(\frac{e^{rt} - 1}{r} \right)^p$$

Für $r \rightarrow 0$ gehen sowohl Nenner, als auch Zähler, gegen 0 (wir können den Grenzwert in die Potenz reinziehen, da die Potenzfunktion stetig ist) und wir können den Satz von L'Hospital anwenden. Das liefert:

$$\lim_{r \rightarrow 0} \left(\frac{e^{rt} - 1}{r} \right)^p = \left(\lim_{r \rightarrow 0} \frac{e^{rt} - 1}{r} \right)^p = \left(\lim_{r \rightarrow 0} \frac{t \cdot e^{rt} - 1}{1} \right)^p = t^p$$

□

Nicht jede Funktion, die durch das additive difference model dargestellt werden kann, ist eine Metrik.

3.2 Segmentally Additive Proximity Structures

Da wir diese Additivität aus der proximity structure herleiten wollen, müssen wir Eigenschaften definieren, aus der man die Additivität herleiten kann. Dazu benötigen wir Folgenden Begriff:

Definition 3.11.

For $a, b, c \in A$, the ternary relation $\langle a, b, c \rangle$ is said to hold iff for all $a', b', c' \in A$:

1. If $(a, b) \succsim (a', b')$ and $(a', c') \succsim (a, c)$, then $(b', c') \succsim (b, c)$
2. If either of the preceding inequalities is strict, then the resulting inequality is also strict.

Further the relation $\langle abc \rangle$ is said to hold if both $\langle a, b, c \rangle$ and $\langle c, b, a \rangle$ hold.

Die Bedingungen, die notwendig sind, um eine Darstellung durch eine Metrik mit additiven Segmenten zu ermöglichen, sind in der folgenden Definition und dem Satz danach zusammengefasst. Der zweite Teil der Definition wird erst später, für das Hauptresultat (Satz 5.13) relevant, im Prinzip geht es darum dass man sich Grenzwerte und Cauchy Folgen anschauen kann, also der normale Begriff von Vollständigkeit.

Definition 3.12 (segmentally additive proximity structure).

A proximity structure $\langle A, \succsim \rangle$, with a ternary relation $\langle \rangle$ defined in terms of $\succsim$, is segmentally additive if and only if the following two axioms hold for all $a, c, e, f \in A$:

1. If $(a, c) \succsim (e, f)$, then there exists $b \in A$ such that $(a, b) \sim (e, f)$ and $\langle abc \rangle$. (segmental solvability)
2. If $e \neq f$, then there exist $b_0, \dots, b_n \in A$, such that $b_0 = a$, $b_n = c$, and $(e, f) \succsim (b_{i-1}, b_i)$ for $i = 1, \dots, n$. (connectedness)

A segmentally additive proximity structure is complete if and only if the following two axioms hold for all $a, c, e, f \in A$:

3. If $e \neq f$, then there exist $b, d \in A$, with $b \neq d$, such that $(e, f) \succ (b, d)$. (Non-discreteness of distances)
4. Let $a_1, \dots, a_n, \dots$ be a sequence of elements of A , such that for all $e, f \in A$, if $e \neq f$, then $(e, f) \succsim (a_i, a_j)$ for all but finitely many pairs (i, j) . Then there exists $b \in A$, such that for all $e, f \in A$, if $e \neq f$, then $(e, f) \succsim (a_i, b)$ for all but finitely many i . (Limits for Cauchy sequences)

4 Additive Difference Models and Structures

4.1 Additive Difference Model

Definition 4.1 (additive difference model).

1. *Decomposability*: The distance between stimuli is a function of componentwise contributions.

$$\delta(a, b) = F[\psi_1(a_1, b_1), \dots, \psi_n(a_n, b_n)], \quad (6)$$

where F is an increasing ¹ real-valued function in each of its n real arguments and ψ_i is a real-valued symmetric function satisfying $\psi_i(a_i, b_i) > \psi_i(a_i, a_i)$ when $a_i \neq b_i$ for all $i = 1, \dots, n$.

2. *Intradimensional subtractivity*: Each componentwise contribution is the absolute value of an appropriate scale difference.

$$\delta(a, b) = F[|\varphi_1(a_1) - \varphi_1(b_1)|, \dots, |\varphi_n(a_n) - \varphi_n(b_n)|], \quad (7)$$

where F is as in eq. (6) and ψ_i is replaced with $|\varphi_i(a_i) - \varphi_i(b_i)|$ for some real-valued function φ_i defined on A_i for $i = 1, \dots, n$.

3. *Interdimensional additivity*: The distance is a function of the sum of componentwise contributions.

$$\delta(a, b) = G \left[\sum_{i=1}^n \psi_i(a_i, b_i) \right], \quad (8)$$

where ψ_i are as in eq. (6) and G is an increasing function in one argument.

4. *Additive difference model*: If both, additivity and subtractivity are assumed, we get the additive difference model.

$$\delta(a, b) = G \left\{ \sum_{i=1}^n F_i[|\varphi_i(a_i) - \varphi_i(b_i)|] \right\}, \quad (9)$$

where F_i and G are all increasing functions in one argument.

4.2 Additive Difference Structure

5 Combining Segmental Additivity with the AD-Model

5.1 Preliminary lemma

Nur kurze zusammenfassung der Notation zur Übersicht Notation:

¹We use the terms increasing and decreasing to denote real-valued functions that are strictly monotonically increasing and decreasing, respectively.

- γ : parametric curve, $L(\gamma)$: length of curve, $L_a^x(\gamma)$: length of subcurve from a to x
- δ_∞ : ∞ -metric, if no subscript then δ is the additive difference metric
- δ : metric, $\delta(a, b)$ metric distance between a and b , $\delta(a) := \delta(0, a)$
- $B_\delta(r) := \{x \in \mathbb{R}^n : \delta(x) < r\}$: the open Ball with radius r around 0 wrt. the metric δ ,
- $B_\delta[r] := \{x \in \mathbb{R}^n : \delta(x) \leq r\}$: the closed Ball with radius r around 0 wrt. the metric δ ,
- $\|x, y\|_\infty := \max_{i \leq n} \{|x_i - y_i|\}$: the maximum metric. We just use ∞ to shorten the notation, e.g. in $B_\infty(r)$.
- $\text{conv}(A)$: convex Hull of A .
- $\text{extreme}(A)$: extreme Points of A .

In this section we will prove the main lemma, which is needed for the proof of the main result. First, we will state the lemma, then give some remarks on the lemma and prove one statement needed for the lemma and then finally prove it. The proof is divided into 5 parts and a whole subsection which is devoted to showing a statement that is needed in the proof. Everything in this section will be assumed as in this Theorem. For example, if we write δ , this implies not any metric but the metric from the lemma with its properties. Moreover all topological terms (open, closed, continuous) are meant with respect to the natural topology of $\mathbb{R}^n$ (i.e. the topology induced by the δ_∞ and the euclidean metric), not the topology induced by δ or the induced topology on S . This will be especially important for the subsection on convexity of the balls. We will begin with the statement:

Lemma 5.1 (Main Lemma).

Let S be an open convex subset of n -dimensional Euclidean space. Let δ be a metric on S that satisfies intradimensional subtractivity² with respect to a continuous function $F : \mathbb{R}_{\geq 0}^n \rightarrow \mathbb{R}_{\geq 0}$ and linear coordinates³ in S . If $\langle S, \delta \rangle$ is a metric with additive segments, then δ is homogeneous, i.e., for any $x, z \in S$ and any real $0 \leq t \leq 1$,

$$\delta[x, x + t(z - x)] = t\delta(x, z).$$

Now we will continue with a remark that makes it possible to move the set such that $0 \in S$ by using the translation invariance of δ .

²i.e. δ takes the form $\delta(x, y) = F(|x_1 - y_1|, \dots, |x_n - y_n|)$.

³Linear coordinates means that the vector representations are cartesian as opposed to, for example, polar, where each coordinate entry is not the value on the corresponding factor of the cartesian product.

Remark 5.2 (Translation invariance and domain of δ and F). (i) δ is translation-invariant, i.e. $\delta(x + y, z + y) = \delta(x, z)$:

$$\begin{aligned}\delta(x + y, z + y) &= F[|x_1 + y_1 - (z_1 + y_1)|, \dots, |x_n + y_n - (z_n + y_n)|] \\ &= F[|x_1 - z_1|, \dots, |x_n - z_n|] \\ &= \delta(x, z)\end{aligned}$$

- (ii) Because δ is invariant to translation in its domain, we can move S and redefine the metric on the shifted set. We do this by defining $S' = \{s - x : s \in S\}$ for $x \in S$ and $\delta'(a, b) := \delta(a + x, b + x)$ for $a, b \in S'$. Since $0 \in S'$, we also can introduce the following shorthand Notation $\delta'(x') := \delta'(0, x')$ for $x' \in S'$. From now on, we will assume that $0 \in S$, because by the argument just provided, we can shift S and δ such that $0 \in S$. Thus we can proceed to proof the statement using the shorthand notation, by which we need to show that:

$$\delta(t \cdot z) = t \cdot \delta(z) \quad \forall t \in [0, 1] \text{ and } \forall z \in S'.$$

- (iii) Since δ is defined by $\delta(x, y) = F(|x_1 - y_1|, \dots, |x_n - y_n|)$ and δ is a metric, $F(x) = 0$ iff $x = 0$ must hold. Moreover the function F has to be defined on a set, where at least the "greatest possible difference" in coordinate i is a possible i -th argument. This means F is defined on a cartesian product of open, real intervals of the form $[0, a_i)$ where

$$a_i = \sup \{|x_i - y_i| : (x_1, \dots, x_n), (y_1, \dots, y_n) \in S\} \in \mathbb{R}_{>0} \cup \{\infty\}.$$

These a_i are these "greatest possible differences" on the i -th coordinate.

- (iv) Moreover, due to the continuity of F and $|\cdot|$, the metric δ is continuous.

Before moving to the proof, we will first show two statements that we will need right at the start. The first one asserts that the distance between a compact and a closed set is greater zero.

Todo: ref?

Lemma 5.3.

If $I \subset \mathbb{R}^n$ is compact, $Y \subset \mathbb{R}^n$ is closed and I and Y are disjoint, then there exists $\varepsilon > 0$ such that $\delta_\infty(p, y) > \varepsilon$ for all $p \in I$ and $y \in Y$.

Proof. We will prove this statement by contradiction. Assume that for all $\varepsilon > 0$ there exists $p \in I$ and $y \in Y$ such that $\delta_\infty(p, y) < \varepsilon$. This means, we can construct two sequences $p_n \in I$ and $y_n \in Y$ such that $\delta_\infty(p_n, y_n) \rightarrow 0$ as $n \rightarrow \infty$. I is compact, thus there exists a convergent subsequence p_{n_m} such that $p_{n_m} \rightarrow p \in I$ for $n \rightarrow \infty$. With the triangle inequality we get

$$\delta_\infty(p, y_{n_m}) \leq \delta_\infty(p, p_{n_m}) + \delta_\infty(p_{n_m}, y_{n_m}) \rightarrow 0 \text{ as } m \rightarrow \infty.$$

Hence, p is a limit point of Y , and since Y is closed, $p \in Y$ must hold which contradicts the assumption. $\square$

The second one allows us to make arbitrary small δ -balls which we will combine with the statement before to assure a distance in δ and not δ_∞ .

Lemma 5.4 (arbitrary small δ -Balls).

Let $\varepsilon > 0$ such that $B_\infty(\varepsilon) \subset S$. Then there exists $\alpha > 0$ such that $B_\delta(\alpha) \subset B_\infty(\varepsilon)$.

Proof. Since δ is a metric defined by $\delta(x, y) = F[|x_1 - y_1|, \dots, |x_n - y_n|]$, we can define the j -th partial function of F by setting all other arguments to zero, i.e. $F^j(u_j) := F(0, \dots, u_j, \dots, 0)$.

F is strictly increasing in every argument implies that F^j is increasing and the continuity of F directly translates to F^j being continuous. Moreover, due to remark 5.2 (iii), $F^j(u_j) = 0$ if and only if $u_j = 0$ holds.

Together, this means that F^j is an increasing, continuous real function with $F^j(0) = 0$.

By setting $\alpha_j = F^j(\varepsilon)$ we get by monotonicity, that $F^j(u_j) < \alpha_j \implies u_j < \varepsilon$. Note, that $F^j(\varepsilon)$ is defined, because $B_\infty(\varepsilon) \subset S'$ implies that ε is a possible distance for each coordinate. (See also remark 5.2 (iii). Using the a_i from the remark, we get that $\varepsilon < a_i$ holds for all i and thus $F^j(\varepsilon)$ is defined for all $j = 1, \dots, n$.)

Doing this for $j = 1, \dots, n$ yields $\alpha_1, \dots, \alpha_n$. By setting $\alpha = \min\{\alpha_1, \dots, \alpha_n\}$ the above holds for every partial function F^j which yields the boundary for the ∞ Metric:

$$F^j(u_j) < \alpha \leq \alpha_j \implies u_j < \varepsilon \quad \forall j = 1, \dots, n \implies \|u\|_\infty < \varepsilon.$$

Using the monotonicity of F we get

$$\delta(u) = F(u) < \alpha \implies F^j(u_j) < \alpha \implies \|u\|_\infty < \varepsilon.$$

□

Now, we will move to the proof of the main lemma. It is suggested to first read the proof to get the main ideas and a big picture understanding, and then read the next section which covers the convexity of spheres. First we will restate the lemma in the form, in which we will prove it.

Lemma.

Let S be an open convex subset of n -dimensional Euclidean space with $0 \in S$. Let δ be a metric on S that satisfies intradimensional subtractivity with respect to linear coordinates in S . If $\langle S, \delta \rangle$ is a metric with additive segments, then δ is homogeneous, i.e., for any $z \in S$ and any real $0 \leq t \leq 1$,

$$\delta(tz) = t\delta(z).$$

The proof consists of 4 parts. First we will find a radius such that all balls with this radius on line from 0 to z are in S . Then we will prove this lemma for $t = \frac{1}{m}$ for $m \in \mathbb{N}$ by proving the two inequalities. After this, we will expand this statement to the rationals and then use continuity to expand it to all real numbers between 0 and 1.

Proof.

Part 0 (Asserting a common distance for all points on the line segment).

Let $z \in S$. Since S is open, $\mathbb{R}^n \setminus S$ is closed. The set $I := \{tz : t \in [0, 1]\}$ is compact, since continuous images $(t \mapsto tz)$ of compact sets $(t \in [0, 1])$ are compact. Todo: ref? Thus, by lemma 5.3 there exists a distance $\varepsilon > 0$ such that $\delta_\infty(p, y) < \varepsilon$ for all $p \in I$ and $y \in \mathbb{R}^n \setminus S$. This means, for all points on the line from 0 to z the ball with radius ε is a subset of S , i.e. for all $t \in [0, 1] : B_\infty(tz, \varepsilon) \subset S$. By lemma 5.4, there exists $\alpha > 0$ such that $B_\delta(\alpha) \subset B_\infty(\alpha)$, and by translation invariance this holds for all points on the line: $B_\delta(\alpha)tz \subset S$ for all $t \in [0, 1]$. Now let $m \in \mathbb{N}$ such that $\frac{1}{m}\delta(z) < \alpha$. In particular, this means that $B_\delta(\alpha)$ and therefore, $B_\delta(\frac{1}{m}\delta(z))$ is bounded (with respect to the ∞ metric).

Part 1 ($t\delta(z) \leq \delta(tz)$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$).

Let $z^{(k)} := \frac{k}{m} \cdot z$ for $k = 0, \dots, m$. $z^{(k)} \in S$ for all k due to convexity of S .

By applying translation invariance, we can calculate the distance between two consecutive points:

$$\delta(z^{(k-1)}, z^{(k)}) = \delta\left(\frac{k-1}{m}z, \frac{k-1}{m}z + \frac{1}{m}z\right) = \delta\left(\frac{1}{m}z\right) \quad \forall k = 1, \dots, m.$$

Applying the triangle inequality $m - 1$ -times and using the above result yields

$$\delta(z) \leq \sum_{k=0}^{m-1} \delta(z^{(k)}, z^{(k+1)}) = m \cdot \delta\left(\frac{1}{m}z\right).$$

Finally, dividing the right side by m yields $\frac{1}{m}\delta(z) \leq \delta(\frac{1}{m}z)$, which concludes the first part.

Part 2 ($t\delta(z) \geq \delta(tz)$ for $t = \frac{1}{m} < d$ with $m \in \mathbb{N}$).

Since δ is a metric with additive segments, there exists a segment from 0 to z with length $\delta(z)$. remark 3.4 yields, that the arc length is a continuous, non-decreasing function. This means, we can divide the segment into m subsegments with length $\frac{1}{m}\delta(z)$ and choose $y^{(0)}, \dots, y^{(m)}$ equally spaced on the segment such that $\delta(y^{(i)}, y^{(i-1)}) = \frac{1}{m}\delta(z)$ and $y^{(0)} = 0$ and $y^{(m)} = z$. Using this, we can define $x^{(i)} := y^{(i)} - y^{(i-1)}$. Translation invariance yields $\delta(x^{(i)}) = \delta(y^{(i)}, y^{(i-1)}) = \frac{1}{m}\delta(z)$ and, due to the choice of m in part 0 $x^{(i)} \in S$ holds. Note, that this is still true, if we increase m .

Further, we can observe that $\frac{1}{m}z$ is a convex combination of $x^{(i)}$:

$$\sum_{i=1}^m \frac{1}{m}x^{(i)} = \sum_{i=1}^m \frac{1}{m}(y^{(1)} - y^{(0)} + y^{(2)} - y^{(1)} + \dots + y^{(m)} - y^{(m-1)}) = \frac{1}{m}(-y^{(0)} + y^{(m)}) = \frac{1}{m}z$$

We showed before, that $\delta(x^{(i)}) = \frac{1}{m}\delta(z)$, which implies $x^{(i)} \in B_\delta[\frac{1}{m}\delta(z)]$. By part 0 $B_\delta(\frac{1}{m}\delta(z))$ is bounded. Therefore we can apply lemma 5.5 from which we get that spheres are convex. This means, that the spheres are closed under convex combination, which implies that $\frac{1}{m}z \in B_\delta[\frac{1}{m}\delta(z)]$ must hold. Thus $\delta(\frac{1}{m}z) \leq \frac{1}{m}\delta(z)$ which concludes the second part of the proof.

Now, by combining part 1 and part 2, we have established the equality for $t = \frac{1}{m}$. Note, that the equality also holds for all $\mathbb{N} \ni m' > m$ since we only needed m to be big enough in part 2 for $x^{(i)}$ to be in S and for convexity of the spheres. Thus, by making m bigger, the crucial points in part 2 and therefore the equality are still true. Next, we will proceed to extending the equality from $t = \frac{1}{m}$ to $t = \frac{k}{m}$ for $k = 0, \dots, m$.

Part 3 ($t \delta(z) = \delta(tz)$ for $t = \frac{k}{m}$ and $t \in \{0, \dots, m\}$).

For $z^{(1)}$ we already have $\delta(0, z^{(1)}) = \frac{1}{m} \delta(0, z)$. Now, for consecutive points, by translation invariance, it holds that

$$\delta(z^{(k-1)}, z^{(k)}) = \delta(z^{(k-2)}, z^{(k-1)}) = \dots = \delta(z^{(1)}, z^{(0)}) = \delta(z^{(1)}, 0) = \frac{1}{m} \delta(0, z).$$

Using this and the triangle inequality, we get

$$\begin{aligned} \delta(0, z) &\leq \delta(0, z^{(1)}) + \delta(z^{(1)}, z^{(2)}) + \dots + \delta(z^{(m-1)}, z^{(m)}) \\ &= n \cdot \frac{1}{n} \delta(0, z) \\ &= \delta(0, z) \\ \implies \delta(0, z) &= \delta(0, z^{(1)}) + \delta(z^{(1)}, z^{(2)}) + \dots + \delta(z^{(m-1)}, z^{(m)}) \end{aligned}$$

For the distance $\delta(0, z^{(2)})$, we get the following two inequalities:

1.

$$\delta(0, z^{(2)}) \leq \delta(0, z^{(1)}) + \delta(z^{(1)}, z^{(2)}) = \frac{2}{m} \delta(0, z)$$

2.

$$\begin{aligned} \delta(0, z) &\leq \delta(0, z^{(2)}) + \delta(z^{(2)}, z) \\ \delta(0, z^{(2)}) &\geq \delta(0, z) - \underbrace{\delta(z^{(2)}, z)}_{\leq \sum_i \delta(z^{(2+i)}, z^{(2+i+1)})} \\ &\geq \delta(0, z) - \sum_{i=0}^{m-3} \delta(z^{(2+i)}, z^{(2+i+1)}) \\ &= \delta(0, z) - \frac{m-2}{m} \delta(0, z^{(k)}) \\ &= \frac{2}{m} \delta(0, z) \end{aligned}$$

These two inequalities yield $\delta(0, z^{(2)}) = \frac{2}{m} \delta(0, z)$. By repeating this for $\delta(0, z^{(k)})$ for every $k = 2, \dots, m-1$, we get the equality for every rational number for m big enough. However every rational number with a denominator smaller than m can be represented by another rational number with denominator greater than m . This yields, that the result is true for every rational t .

Part 4 ($t\delta(z) = \delta(tz)$ for $t \in [0, 1]$).

The final step can be completed using continuity of the metric. Now let $t \in [0, 1]$ and $t_n \in \mathbb{Q}$ with $t_n \rightarrow t$ for $n \rightarrow \infty$. Then

$$\begin{aligned}\delta(tz) &= \delta\left(\lim_{n \rightarrow \infty} t_n z\right) \\ &= \lim_{n \rightarrow \infty} \delta(t_n z) && \text{(continuity)} \\ &= \lim_{n \rightarrow \infty} t_n \delta(z) && (t_n \in \mathbb{Q} \text{ and part 3}) \\ &= t \delta(z)\end{aligned}$$

□

5.1.1 Convexity of δ Balls

In this subsection we will proof the following lemma which is needed to conclude the proof of lemma 5.1. Everything will be assumed as in the lemma 5.1. Note, that all topological terms are with respect to the natural topology of the euclidean space.

Lemma 5.5 (Convexity of δ -Spheres).

If $B_\delta[\alpha] \subset S$ is bounded, then it is convex.

First, we will state some definitions and results on basic convex geometry that we need, and prove the theorem at the end of the section. The definitions and results on convex geometry are taken out of (Hug & Weil, 2020).

First of all, we need the notions of convex combination, the convex hull and extreme points. A convex combination is similar to a linear combination, but has the additional restraint that the coefficients must be positive and add up to 1. The convex hull of a set is, like any other hulls, the smallest convex set that contains the set.

Definition 5.6 (Convex Combination and Hull).

1. Let $k \in \mathbb{N}$, let $x_1, \dots, x_k \in \mathbb{R}^n$, and let $a_1, \dots, a_k \in [0, 1]$ be such that $a_1 + \dots + a_k = 1$. Then $a_1 x_1 + \dots + a_k x_k$ is called a *convex combination* of the points $x_1, \dots, x_k$.
2. For $A \subset \mathbb{R}^n$ the *convex Hull* $\text{conv}(A)$ is the smallest convex set containing A , i.e.

$$\text{conv}(A) := \cap \{C \subset \mathbb{R}^n : A \subset C, C \text{ convex}\}$$

The next theorem sums up the relationship between convex combination and convex hulls.

Theorem 5.7.

For $A \subset \mathbb{R}^n$ the convex Hull is the set of all convex combination of points in A :

$$\text{conv}(A) = \left\{ \sum_{i=1}^k a_i x_i : k \in \mathbb{N}, x_1, \dots, x_k \in A, a_1, \dots, a_k \in [0, 1] : \sum_{i=1}^k a_i = 1 \right\}$$

Moreover, the convex hull preserves compactness, as the following theorem shows.

Theorem 5.8 (convex Hull of compact set is compact).

If $A \subset \mathbb{R}^n$ is compact, then $\text{conv}(A)$ is compact.

Next, we will introduce extreme points, which are, in a sense, characteristic points of a set that cannot be represented by a (strict) convex combination.

Definition 5.9 (Extreme Point).

Let $A \subset \mathbb{R}^n$ be closed and convex. A point $x \in A$ is called an *extreme point* of A if x cannot be represented as a nontrivial (or strict) convex combination of points of A , that is, if $x = a \cdot y + (1 - a) \cdot z$ with $y, z \in A$ and $a \in (0, 1)$, implies that $x = y = z$. The set of all extreme points of A is denoted by $\text{extreme}(A)$.

For compact and convex sets, the extreme points are enough to "build" the convex hull of the set by doing convex combinations. In particular, this means that for a compact and convex set, every point can be expressed as a convex combination of extreme points.

Theorem 5.10 (Minkowski Theorem).

Let $A \subset \mathbb{R}^n$ be compact and convex. Then $A = \text{conv}(\text{extreme } A)$.

Before moving to the convexity of balls, we will first prove two lemma, which we will need while proving convexity of spheres. The first one is the openness/closedness of δ -balls.

Lemma 5.11 (δ -Balls are open/closed).

For every radius $r > 0$, the δ -Balls are open ($B_\delta(r)$) or closed ($B_\delta[r]$).

Proof. $\delta : S' \times S' \rightarrow \mathbb{R}_{\geq 0}$ is continuous with $S' \subseteq \mathbb{R}^n$ and therefore $y \mapsto \delta(y) = \delta(0, y)$ is continuous.

Consider the closed interval $U' = [0, \alpha] \subset \mathbb{R}_{\geq 0}$. Then

$$\begin{aligned} \delta^{-1}(U') &= \{y \in S' : \delta(y) \in U'\} \\ &= \{y \in S' : \delta(0, y) \leq \alpha\} \\ &= B_\delta[\alpha] \end{aligned}$$

Due to topological continuity of δ (which is equivalent to preimages of closed sets being closed) $B_\delta[\alpha]$ is closed. By translation invariance this holds for closed balls with any center in S . **Todo: ref topo stetig?** Moreover, we can obtain the same result for open balls by repeating this proof and replacing the closed interval with an open one and $\leq$ with $<$. $\square$

The second lemma shows, that extreme points of balls get compressed by a factor of $\alpha \in \mathbb{N}$, if the radius of the ball gets shrunk by the same factor.

Lemma 5.12 (compressing extreme points).

Let $\alpha \in \mathbb{R}_{>0}$ and $r \in \mathbb{N}_{>0}$. Further let $E(\alpha) := \text{extreme}(B_\delta[\alpha])$ and $C[\alpha] := \text{conv}(B_\delta[\alpha])$. Then, $y \in E(\alpha)$ implies $\frac{1}{r}y \in E(\frac{\alpha}{r})$.

Proof. We will prove this statement by contraposition.

Assume, that $\frac{1}{r}y$ is not an extreme point, i.e. $\frac{1}{r}y \in C\left[\frac{\alpha}{r}\right] \setminus E\left(\frac{\alpha}{r}\right)$. Then we can write $\frac{1}{r}y$ as a strict convex combination of elements $w^i \in C\left[\frac{\alpha}{r}\right]$:

$$\frac{1}{r}y = \sum_{i=1}^p \lambda_i w^i \quad \text{with} \quad \sum_{i=1}^p \lambda_i = 1, \quad \lambda_i > 0, \quad w^i \in C\left[\frac{\alpha}{r}\right] \quad \text{and} \quad p \in \mathbb{N}$$

But because $C\left[\frac{\alpha}{r}\right] = \text{conv}(B_\delta\left[\frac{\alpha}{r}\right])$ holds, we can write any element from $C\left[\frac{\alpha}{r}\right]$ as a convex combination of elements from $B_\delta\left[\frac{\alpha}{r}\right]$ and therefore we can assume $w^{(i)} \in B_\delta\left[\frac{\alpha}{r}\right]$.

By translation invariance and the triangle inequality we get for all i :

$$\begin{aligned} \delta\left(rw^{(i)}\right) &\leq \delta\left(0, w^{(i)}\right) + \delta\left(w^{(i)}, rw^{(i)}\right) && \text{triangle inequality} \\ &= \delta\left(w^{(i)}\right) + \delta\left(0, (r-1) \cdot w^{(i)}\right) && \text{translation invariance} \\ &\leq \delta\left(w^{(i)}\right) + \delta\left(0, w^{(i)}\right) + \delta\left(w^{(i)}, (r-1) \cdot w^{(i)}\right) && \text{triangle inequality} \\ &= 2 \cdot \delta\left(w^{(i)}\right) + \delta\left(0, (r-2) \cdot w^{(i)}\right) && \text{translation invariance} \\ &\dots \\ &\leq r \cdot \delta\left(w^{(i)}\right) \\ &= r \cdot \frac{\alpha}{r} = \alpha \\ \implies rw^{(i)} &\in B_\delta[\alpha] \end{aligned}$$

This yields, that $y = \sum_{i=1}^p \lambda_i rw^{(i)}$ is a strict convex combination (because $\lambda_i r \neq 0$) and therefore y is not in $E(\alpha)$. $\square$

Now we can prove the convexity of δ -balls.

Todo: ref heine-borel?

Proof of lemma 5.5. Let $E(\alpha) := \text{extreme}(B_\delta[\alpha])$ and $C[\alpha] := \text{conv}(B_\delta[\alpha])$. By lemma 5.11 $B_\delta[\alpha]$ is closed and by assumption it is also bounded. Therefore, from the Heine-Borel Theorem, it follows that $B_\delta[\alpha]$ is compact.

By definition and theorem 5.8 it follows, that $C[\alpha]$ is compact (and therefore closed, which is the reason why we denote it with square brackets) and convex. Application of the Minkowski Theorem (theorem 5.10) yields:

$$C[\alpha] = \text{conv}(\text{extreme}(C[\alpha]))$$

Further, it holds that $\text{extreme}(C[\alpha]) = \text{extreme}(B_\delta[\alpha])$ (By doing convex combinations, no new extreme points can be introduced because these are the points, which cannot be represented by a convex combination). Combining this and the statement above yields:

$$B_\delta[\alpha] \subset C[\alpha] = \text{conv}(\text{extreme}(B_\delta[\alpha])).$$

Now, if we prove that $\text{conv}(\text{extreme}(B_\delta[\alpha])) \subset B_\delta[\alpha]$, we get that $B_\delta[\alpha] = C[\alpha]$ which proves the convexity of $B_\delta[\alpha]$. We simplify this statement before proving it.

$$\begin{aligned}
& \text{conv}(\underbrace{\text{extreme}(B_\delta[\alpha])}_{E(\alpha)}) \subset B \\
& \quad \Updownarrow \\
& \sum_{i=1}^p \lambda_i y^{(i)} \in B_\delta[\alpha], \quad p \in \mathbb{N}_{>0}, \begin{cases} y^{(i)} \in E(\alpha) \\ \lambda_i \in \mathbb{R}_{\geq 0}, \sum_{i=1}^p \lambda_i = 1 \end{cases} \quad (\text{Definition}) \\
& \quad \Updownarrow \\
& \sum_{i=1}^p \lambda_i y^{(i)} \in B_\delta[\alpha], \quad p \in \mathbb{N}_{>0}, \begin{cases} y^{(i)} \in E(\alpha) \\ \lambda_i \in \mathbb{Q}_{\geq 0}, \sum_{i=1}^p \lambda_i = 1 \end{cases} \quad (B \text{ closed, } \mathbb{Q} \text{ dense in } \mathbb{R}^4) \\
& \quad \Updownarrow \\
& \sum_{i=1}^r \frac{1}{r} y^{(i)} \in B_\delta[\alpha], \quad r \in \mathbb{N}_{>0}, y^{(i)} \in E(\alpha) \quad (r \text{ common denominator})
\end{aligned}$$

Thus, we will continue proving that for $r \in \mathbb{R}_{>0}$ and $y^{(i)} \in E(\alpha)$ it holds that $\sum_{i=1}^r \frac{1}{r} y^{(i)} \in B_\delta[\alpha]$.

For $y^{(i)} \in E(\alpha)$ it holds by lemma 5.12 that $\frac{1}{r} y^{(i)} \in E(\frac{\alpha}{r}) \subset B_\delta[\frac{\alpha}{r}]$. This means, that for the metric distance of the convex combination the following holds:

$$\begin{aligned}
\delta\left(0, \sum_{i=1}^r \frac{1}{r} y^{(i)}\right) &\leq \delta\left(0, \frac{1}{r} y^{(1)}\right) + \delta\left(\frac{1}{r}, \sum_{i=1}^r \frac{1}{r} y^{(i)}\right) && (\text{triangle inequality}) \\
&= \delta\left(0, \frac{1}{r} y^{(1)}\right) + \delta\left(0, \sum_{i=2}^r \frac{1}{r} y^{(i)}\right) && (\text{translation invariance}) \\
&\leq \frac{\alpha}{r} + \delta\left(0, \sum_{i=2}^r \frac{1}{r} y^{(i)}\right) && (\text{translation invariance}) \\
&\dots && (\text{repeat}) \\
&\leq r \frac{\alpha}{r} = \alpha
\end{aligned}$$

This yields that $\sum_{i=1}^r \frac{1}{r} y^{(i)} \in B_\delta[\alpha]$ holds which proves the statement. $\square$

⁴Density implies, that we can express every real number as a limit of rational numbers. B is closed, and therefore all limits of rational coefficients are included, if the statement holds for rational numbers.

Theorem 5.13 (Eindeutigkeitssatz zu l^p Metriken).

Sei $\langle A, \succsim \rangle$ eine vollständige, segmentally additive proximity structure, die auch eine additive difference structure ist.

Ferner sei δ homogen, also $\delta(x, x + t(z - x)) = t\delta(x, z)$, für $t > 0$.

Dann gibt es ein eindeutiges $r \geq 1$ und reellwertige Funktionen φ_i , die auf A definiert sind, $i = 1, \dots, n$, die bis auf eine positive lineare Transformation eindeutig sind, so dass für alle $a, b, c, d \in A$,

$$(a, b) \succsim (c, d) \iff \delta(a, b) \geq \delta(c, d),$$

wobei

$$\delta(a, b) = \left[\sum_{i=1}^n |\varphi_i(a) - \varphi_i(b)|^r \right]^{1/r}.$$

Remark 5.14. Dieser Satz ist ursprünglich ohne die Zusatzannahme der Homogenität von δ formuliert. Jedoch konnten wir den Beweis an einer Stelle nicht nachvollziehen bzw. denken dass er möglicherweise sogar falsch ist, und nehmen daher zusätzlich diese Eigenschaft an.

Proof. Der Beweis passiert in mehreren Schritten. Zuerst sagen wir etwas über die Definitions- und Zielbereiche der Funktionen F_i und finden damit Grenzen für den Definitionsbereich von F . Danach argumentieren wir, dass wir F bis auf die positiven reellen Zahlen $[0, \infty)$ erweitern können und dass diese Erweiterung eine Funktionsgleichung impliziert, die nur von $f(x) = kx^p$ erfüllt wird. Im letzten Schritt zeigen wir, dass F auf ganz $\mathbb{R}^+$ gleich der Erweiterung sein muss und damit δ eine l^p Metrik sein muss.

Nach Satz [Todo: ref hinr-notw-beds](#) liefert das additive difference Model

$$\delta(a, b) = G \left(\sum_{i=1}^n F_i(|\varphi_i(a_i) - \varphi_i(b_i)|) \right)$$

nur dann eine Metrik, wenn $F_i(\alpha) = F(t_i \alpha)$ und $F = G^{-1}$ gilt. Daher ist F^{-1} für alle Zahlen $\sum_{i=1}^n F_i(|\varphi_i(a_i) - \varphi_i(b_i)|)$ definiert, und jedes F_i stimmt mit F auf seinen Definitionsbereich überein (bis auf den Faktor t_i , der das Argument multipliziert und den Definitionsbereich anpasst).

Da die Definitionsbereiche von F_i Intervalle sind (weil δ eine Metrik mit additiven Segmenten ist), können wir $\omega > 0$ wählen, so dass für jedes $i = 1, \dots, n$ und jedes α mit $0 \leq \alpha \leq \omega/t_i$ gilt, dass α im Bereich von F_i liegt.

Insbesondere, wenn wir $\omega_i = \omega/t_i$ setzen, ist $F^{-1}[\sum_{i=1}^n F_i(\omega_i)] = F^{-1}[nF(\omega)] =: \Omega$ definiert. Damit ist $F = G^{-1}$ mindestens im Intervall $[0, \Omega]$ definiert. Wir können F nun wie folgt erweitern.

Für $\alpha < \omega$ gilt mit der Homogenität

$$\begin{aligned} F^{-1}[nF(\alpha)] &= F^{-1}[nF(\alpha\omega/\omega)] \\ &= (\alpha/\omega)F^{-1}[nF(\omega)] \\ &= \alpha\Omega/\omega. \end{aligned}$$

Das liefert durch Einsetzen der korrekten Werte für α ($\tilde{\alpha} = \alpha \frac{\omega}{\Omega}$ und $\tilde{\alpha} = F^{-1}(\frac{\alpha}{n})$)

$$F(a) = nF(a\omega/\Omega), \quad 0 \leq a \leq \Omega; \quad (10)$$

$$F^{-1}(a) = (\Omega/\omega)F^{-1}(a/n), \quad 0 \leq a \leq nF(\omega). \quad (11)$$

Diese beiden Gleichungen können wir jetzt nutzen, um den Definitionsbereich von F und F^{-1} zu erweitern. Wir nutzen (Todo: ref f-erw) indem wir F durch die rechte Seite definieren (für Werte, bei denen die rechte Seite definiert ist, also für $0 \leq a \leq \Omega^2/\omega$). Da $\Omega^2/\omega > \Omega$ gilt erweitert dies F wirklich über das Intervall $[0, \Omega]$ hinaus.

Entsprechend erfüllt F^{-1} Gleichung (Todo: ref f-1-erw) für das erweiterte Intervall $0 \leq a \leq nF(\Omega) = n^2F(\omega)$.

Darüber hinaus gilt die Homogenität nun für $0 \leq t\alpha_i \leq \Omega$:

$$\begin{aligned} F^{-1} \left[\sum_{i=1}^n F(t\alpha_i) \right] &= F^{-1} \left[\sum_{i=1}^n nF(t\alpha_i\omega \setminus \Omega) \right] \quad (\text{nach Gleichung Todo: ref f-erw }) \\ &= F^{-1} \left[n \sum_{i=1}^n F(t\alpha_i\omega \setminus \Omega) \right] \\ &= \frac{\Omega}{\omega} F^{-1} \left[\sum_{i=1}^n F(t\alpha_i\omega \setminus \Omega) \right] \quad (\text{nach Gleichung Todo: ref f-1-erw }) \\ &= \frac{\Omega}{\omega} t F^{-1} \left[\sum_{i=1}^n F(\alpha_i\omega \setminus \Omega) \right] \quad (\text{wegen Homogenität, da } \alpha_i\omega \setminus \Omega \leq \omega) \\ &= t F^{-1} \left[n \sum_{i=1}^n F(\alpha_i\omega \setminus \Omega) \right] \quad (\text{nach Gleichung Todo: ref f-1-erw }) \\ &= t F^{-1} \left[\sum_{i=1}^n nF(\alpha_i\omega \setminus \Omega) \right] \\ &= t F^{-1} \left[\sum_{i=1}^n F(\alpha_i) \right] \quad (\text{nach Gleichung Todo: ref f-erw }) \end{aligned}$$

Durch wiederholte Anwendung erweitern sich die Intervalle, in denen F definiert ist und Homogenität gültig ist, jedes Mal um einen Faktor Ω/ω . Somit können wir F auf $[0, \infty)$ erweitern sodass die Homogenität erhalten bleibt.

Die Funktion, die durch das erweitern entstanden ist, muss somit mindestens auf dem Intervall $[0, \infty)$ mit der ursprünglichen Funktion F übereinstimmen. Außerdem gilt mit Gleichung (Todo: ref f-1-erw) durch Einsetzen der Mittelwertfunktion $F^{-1} \left[\left(\frac{1}{n} \sum_{i=1}^n F(\alpha_i) \right) \right]$:

$$F^{-1} \left[\frac{1}{n} \sum_{i=1}^n F(t\alpha_i) \right] = t F^{-1} \left[\frac{1}{n} \sum_{i=1}^n F(\alpha_i) \right].$$

Nach **hardy1952inequalities** sind die einzigen monotonen Funktionen, die diese Gleichung für $0 \leq \alpha < \infty$ erfüllen, Funktionen der Form $k\alpha^r$ (siehe auch **aczel1966lectures**).

Damit gilt, dass $F(\alpha) = k\alpha^r$ mindestens auf dem Intervall $[0, \Omega]$ gelten muss. Aber jetzt können wir, für jedes $|\varphi_i(a_i) - \varphi_i(b_i)|$, ein $t > 0$ so klein finden, dass $t|\varphi_i(a_i) - \varphi_i(b_i)| < \omega, i = 1, \dots, n$. Anwenden von der Monotonie liefert (wir können annehmen, dass $t \leq 1$)

$$\begin{aligned}\delta(a, b) &= F^{-1} \left(\sum_{i=1}^n F(|\varphi_i(a_i) - \varphi_i(b_i)|) \right) \\ &= \frac{1}{t} F^{-1} \left(\sum_{i=1}^n F(t|\varphi_i(a_i) - \varphi_i(b_i)|) \right) \\ &= \frac{1}{t} \left(\sum_{i=1}^n (t|\varphi_i(a_i) - \varphi_i(b_i)|)^r \right)^{1/r} \quad \text{weil } t|\varphi_i(a_i) - \varphi_i(b_i)| \leq \omega \\ &= \left(\sum_{i=1}^n |\varphi_i(a_i) - \varphi_i(b_i)|^r \right)^{1/r}\end{aligned}$$

Damit ist δ eine l^p Metrik, und durch Satz **Todo: ref add-diff-prox-struct-repr** ist $F = k \cdot \alpha^r$ und damit $F_i = k(t_i \alpha)^r$ oder $F_i(\alpha) = k\alpha^r$, wenn wir die t_i in die φ_i einbauen. Schließlich folgt die Einschränkung $1 \leq r < \infty$ aus der Dreiecksungleichung oder, genauer gesagt, aus der Minkowski-Ungleichung. □

6 Discussion

References

- Busemann, H. (1955). *The geometry of geodesics* (Vol. 6).
Hug, D., & Weil, W. (2020). *Lectures on Convex Geometry* (Vol. 286). Springer International Publishing. <https://doi.org/10.1007/978-3-030-50180-8>